MINI PROJECT

“Payment System Trends in India”

PROCESS DOCUMENTATION

DATA CLEANING

- Replace the null RTGS Value with “0”, since the null value corresponds to days of bank holiday and hence no RTGS transactions.
- Dropping the data for data dates after 31-05-2025, since the data was inconsistent, data for the month of June was not available and duplicate values were observed.
- Replace the null NACH Value with “0”, since the null value corresponds to days of bank holiday and hence no NACH transactions till July 2021.
- Replace the null CTS Value with “0”, since the null value corresponds to days of bank holiday and hence no CTS transactions.
- Card transaction data was dropped since the data for a major period of the analysis was not captured for in the source data.
- The text was standardised and inconsistencies were removed.
- The data was segregated into different segments “Digital Payments”, “Banking Payments” and “ATM/CASH”. The data was segmented using XLOOKUP formula.
- A few columns were dropped, as they were not directly linked to study.
- Final count of rows and columns:
  COLUMNS – 26
  ROWS – 1826

DATA TRANSFORMATION

- The columns of the ‘Clean Data’ were imported in Power BI, and the columns except the date column were Unpivoted.
- The Payment Channel was split in two in order to get the ‘Payment Metric’ separately.
- The segmented data was also imported to the power query.
- A new table was formed in the name of ‘Channels’, wherein the channel id, payment channel and payment category were recorded.
- The channel id and payment category were added in the table ‘Payment Data’ using ‘Merge Queries’.
- The steps 1 and 2 were also performed in the segmented data table.

DAX QUERIES

The following DAX queries were performed:

- `Total Payment Value = CALCULATE(SUM('PAYMENT DATA'[PAYMENT VALUE]), 'PAYMENT DATA'[PAYMENT METRIC]="VALUE")`
- `Total Payment Volume = CALCULATE(SUM('PAYMENT DATA'[PAYMENT VALUE]),'PAYMENT DATA'[PAYMENT METRIC]="VOLUME")`
- `Average Transaction Value = DIVIDE([Total Payment Value],[Total Payment Volume])`
- `Digital Payment Share = DIVIDE('DIGITAL PAYMENTS'[Total Digital Payments Value],[Total Payment Value])`
- `Banking Payment Share = DIVIDE('BANKING PAYMENTS'[Total Banking Payments Value],[Total Payment Value])`
- `ATM/Cash Payment Share = DIVIDE('ATMCASH'[Total ATM/CASH Payments Value],[Total Payment Value])`
- `Number of Channels = DISTINCTCOUNT('PAYMENT DATA'[PAYMNET CHANNEL])`
- `Quarter Number = "Q" & FORMAT('PAYMENT DATA'[Date], "Q")`
- `Quarter Year = "Q" & FORMAT('PAYMENT DATA'[Date], "Q")& " " & FORMAT('PAYMENT DATA'[Date],"YYYY")`
- `Total Digital Payments Value = CALCULATE(SUM('DIGITAL PAYMENTS'[PAYMENT VALUE]),'DIGITAL PAYMENTS'[PAYMENT METRIC]= "VALUE")`
- `Total Banking Payments Value = CALCULATE(SUM('BANKING PAYMENTS'[PAYMENT VALUE]), 'BANKING PAYMENTS'[PAYMENT METRIC]="VALUE")`
- `Total ATM/CASH Payments Value = CALCULATE(SUM('ATMCASH'[PAYMENT VALUE]),'ATMCASH'[PAYMENT METRIC]="VALUE")`

MODEL VIEW

The model relationship was obtained by relating the ‘Channel id’ column of the ‘Channels Table’ with the other tables. Hence obtaining a one-to-many relationship.